UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	:	COMPUTER SECURITY
COURSE CODE	:	CSC662
EXAMINATION	:	OCTOBER 2026 (MOCK)
TIME	:	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of two (2) parts: PART A (10 Questions)
   PART B (2 Questions)

2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.

3. Do not bring any material into the examination room unless permission is given by the invigilator.

4. Please check to make sure that this examination pack consists of:

   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty

5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 4 printed pages

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PART A (80 MARKS)

QUESTION 1

a) Explain the difference between morals, ethics and laws.

(2 marks)

b) Explain six requirements that a professional code of ethics and conduct, such as the ACM code, places on computing professionals.

(6 marks)

QUESTION 2

a) List two characteristics of Trojan horses as described in the lecture.

(2 marks)

b) Refer to your answer in (a). Explain six techniques by which malware infects programs or propagates, and the impact of the hidden components it installs.

(6 marks)

QUESTION 3

a) List two formal security models named in the lecture and state the focus of each.

(2 marks)

b) Describe six aspects of access control structures, covering the access control matrix, access control lists and capabilities, and the requirements such structures must satisfy.

(6 marks)

QUESTION 4

a) State the impact of a disclosure threat and the impact of a disruption threat.

(2 marks)

b) Discuss the Samy worm as a case study of XSS malware by describing six aspects of how it operated and spread.

(6 marks)

QUESTION 5

a) Explain how a chain of trust is established in Trusted Computing using the TPM.

(2 marks)

b) Explain six components and functions of the NGSCB architecture, covering both the Nexus and the Nexus Computing Agent.

(6 marks)

QUESTION 6

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a) Explain how NIDS sensors should be positioned in a network and why.

(2 marks)

a) Discuss six preventive network security mechanisms that an organisation can deploy, explaining the protective role of each mechanism.

(6 marks)

QUESTION 7

a) Explain two limitations of passive intrusion detection systems in handling attacks.

(2 marks)

b) Compare a threat and an attack as security concepts, and contrast how misuse-based and anomaly-based detection models each counter them, noting one strength and one weakness of each model.

(6 marks)

QUESTION 8

a) State the two stages of the biometric authentication process.

(2 marks)

b) Explain the categories of biometric technologies and the practical issues that affect their use in authentication.

(6 marks)

QUESTION 9

a) State the four goals of access control and the role of a reference monitor.

(2 marks)

b) State six aspects of the Chinese Wall formal security model, covering its purpose, its simple security rule, its write rule, and how its permissions behave.

(6 marks)

QUESTION 10

a) Explain why buffer overflows arise in software and how they can be exploited.

(2 marks)

b) State six Windows security mechanisms or components and, for each, what it provides.

(6 marks)

PART B (20 MARKS)

QUESTION 1

A small company runs a shared Linux server where staff log in with user names and passwords. After several accounts were guessed by an attacker, the administrator is asked

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to strengthen how passwords are protected and managed on the system.

a) Explain five password security controls that the administrator can apply to protect the server's user accounts.

(5 marks)

b) State five points about single sign-on and Kerberos that the administrator in the scenario could use to reduce staff reliance on typed passwords on the shared server.

(5 marks)

QUESTION 2

a) Discuss the ethical impacts of technology by examining how professional codes of ethics guide computing professionals, and evaluate the limitations of such codes in addressing ethical concerns.

(5 marks)

b) Explain five ways an organization can demonstrate responsibility for privacy when handling personal information, referring to the ethical dimensions of information use.

(5 marks)

END OF QUESTION PAPER

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